

YELLOWSTONE REGIONAL, WY, USA

WMO: 726700

Lat: 44.517N

Lon: 109.017W

Elev: 5092

StdP: 12.19

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24045

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-3.5	-19.7	2.3	-5.2	-12.6	3.3	-1.6	27.5	37.0	25.5	37.6	4.5	30	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	90.5	59.3	88.0	58.6	84.1	57.8	62.2	82.3	60.9	81.2	59.6	79.6	9.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.1	78.1	68.7	53.8	74.6	68.3	51.8	69.2	67.6	30.5	81.1	29.4	81.3	28.5	79.9	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.6	20.2	18.1	DB	-16.5	95.6	5.7	2.6	-20.6	97.5	-23.9	99.0	-27.1	100.5	-31.2	102.4	
			WB	-17.0	64.9	5.0	1.7	-20.6	66.1	-23.5	67.1	-26.4	68.1	-30.0	69.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	26.3	27.5	36.8	43.0	52.2	61.7	70.8	68.6	59.0	45.9	34.7	26.5
	DBStd	18.62	12.84	13.07	11.37	9.42	8.59	7.50	5.81	6.26	9.23	10.35	12.47	12.15
	HDD50	3541	734	632	422	246	77	7	0	0	27	198	471	727
	HDD65	7306	1199	1051	874	659	404	147	18	38	222	594	908	1192
	CDD50	2156	0	1	12	37	145	360	645	576	296	71	13	0
	CDD65	445	0	0	0	0	6	49	198	149	41	2	0	0
	CDH74	5186	0	0	0	8	106	637	2241	1669	499	26	0	0
	CDH80	1854	0	0	0	0	14	193	900	608	137	2	0	0
(14) Wind	WSAvg	7.6	7.8	8.0	8.6	8.5	7.7	7.4	6.9	6.6	6.8	7.4	7.9	7.9
(15) Precipitation	PrecAvg	9.1	0.4	0.2	0.4	0.9	1.8	1.6	0.9	0.8	1.0	0.7	0.4	0.3
	PrecMax	12.4	1.9	1.2	1.1	2.3	4.7	3.5	3.3	2.6	3.3	2.8	0.8	1.2
	PrecMin	5.8	0.0	0.0	0.0	0.1	0.1	0.1	0.2	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.5	0.2	0.2	0.6	1.1	0.9	0.7	0.6	0.8	0.6	0.2	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	58.9	67.4	74.5	81.7	90.5	95.4	93.0	88.4	77.5	64.8	54.5
		MCWB	40.6	41.0	45.9	49.9	54.8	58.7	60.2	58.9	58.1	52.0	45.4	40.1
	2%	DB	50.5	54.0	62.8	69.6	76.9	85.4	91.1	89.8	83.8	71.9	60.9	51.4
		MCWB	37.9	39.3	43.2	47.2	53.4	57.3	59.9	58.4	56.6	49.5	43.8	39.0
	5%	DB	46.5	48.9	57.4	63.9	72.6	81.4	88.5	86.4	80.7	66.2	56.1	46.5
		MCWB	35.5	37.5	41.1	45.5	51.6	57.4	59.6	57.6	54.8	47.6	41.3	36.1
	10%	DB	44.8	45.3	54.1	59.8	68.5	77.1	84.6	83.1	75.2	62.7	53.0	43.7
		MCWB	34.3	34.6	39.9	43.8	50.6	56.2	58.8	57.3	53.1	45.9	40.0	33.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	42.4	46.4	51.0	57.9	63.1	64.6	62.8	61.1	53.5	47.6	41.5
		MCDB	52.4	55.7	65.0	71.6	74.6	80.8	83.0	81.7	79.4	72.6	62.3	52.1
	2%	WB	38.7	40.0	44.0	48.4	55.2	60.8	62.8	60.9	58.5	50.8	44.4	39.2
		MCDB	49.2	52.3	60.6	65.7	72.4	78.7	83.4	81.3	78.4	68.5	59.1	49.0
	5%	WB	36.4	37.6	42.0	46.4	53.3	58.9	61.5	59.7	56.3	48.6	42.1	36.7
		MCDB	46.3	48.4	56.8	62.5	70.0	76.8	82.7	80.1	76.1	64.2	55.3	46.4
	10%	WB	34.3	35.1	40.1	44.3	51.2	57.1	60.3	58.6	54.3	46.5	40.0	34.0
		MCDB	43.9	44.8	53.0	58.5	66.4	74.6	81.0	78.8	73.2	60.6	51.9	43.4
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	19.5	20.9	20.6	21.0	23.4	25.7	25.2	23.3	20.7	18.9	18.6
		MCDBR	21.3	22.7	25.3	27.4	28.0	29.2	29.2	29.8	28.8	27.0	22.0	20.1
		MCWBR	13.2	13.8	13.4	13.5	12.4	12.6	11.4	11.5	12.4	13.2	12.2	12.7
	5% WB	MCDBR	21.0	22.1	24.4	25.6	25.2	26.4	26.3	25.8	26.1	25.1	21.4	19.9
		MCWBR	13.7	13.9	13.3	13.1	11.9	12.4	11.1	11.0	11.8	13.2	12.5	13.1
(40) Clear-Sky Solar Irradiance	taub		0.235	0.242	0.258	0.294	0.318	0.323	0.340	0.338	0.306	0.273	0.247	0.232
	taud		2.512	2.518	2.519	2.440	2.447	2.471	2.453	2.435	2.525	2.585	2.565	2.519
	Ebn at Noon		293	309	315	308	302	299	292	290	293	292	285	283
	Edh at Noon		21	25	29	34	35	34	34	34	28	23	19	18
(44) All-Sky Solar Radiation	RadAvg		571	884	1310	1676	1869	2190	2229	1880	1502	962	613	454
	RadStd		45	63	102	123	174	158	111	86	113	75	54	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page